

# MARTINA ALUTTO

***Postdoctoral Researcher***
***at KTH Royal Institute of Technology***

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## RESEARCH EXPERIENCE

**Postdoctoral Researcher** **2025 – current**
*KTH Royal Institute of Technology, Stockholm, Sweden*
*Supervisors:* ass. prof. Angela Fontan and prof. Karl Henrik Johansson
*Research topics:* Social networks, opinion dynamics, network systems
Affiliated with the Wallenberg AI, Autonomous Systems and Software Program (WASP).

**Research Fellow** **2024 – 2025**
*National Research Council (CNR), Torino, Italy*
*Supervisors:* prof. Chiara Ravazzi and prof. Fabrizio Dabbene
Analysis and control of innovation adoption models coupled with opinion dynamics.

**PhD Student in Pure and Applied Mathematics** **2021 – 2024**
*Politecnico di Torino – Università di Torino, Torino, Italy*
*Supervisors:* prof. Fabio Fagnani and prof. Giacomo Como
Thesis: *Network Effects and Behavioral Feedback in Epidemic Models*

## RESEARCH VISITS

**University of Tokyo, Tokyo, Japan** **06/2025 – 07/2025**
*Supervisor:* prof. Hideaki Ishii
Control of innovation adoption-opinion models (ASPIRE program).

**Kyushu University, Fukuoka, Japan** **07/2025**
*Supervisor:* prof. Yoshio Ebihara

**Cornell University, Ithaca (NY), USA** **09/2023 – 02/2024**
*Supervisor:* prof. Francesca Parise
Optimal control problems on epidemic models.

## EDUCATION

**Ph.D. in Pure and Applied Mathematics** **2021 – 2025**
*Politecnico di Torino – Università di Torino, Torino, Italy*
Final evaluation: cum laude
Thesis: *"Network Effects and Behavioral Feedback in Epidemic Models"*
Supervisors: prof. Fabio Fagnani and prof. Giacomo Como

**M.Sc. in Mathematical Engineering** **2018 – 2021**
*Politecnico di Torino, Torino, Italy — Final grade: 110/110 cum laude*
Thesis: *"Feedback Control Policies and Network Effects in Epidemics Models"*
Thesis supervisors: prof. Fabio Fagnani and prof. Giacomo Como

**B.Sc. in Mathematics for Engineering** **2015 – 2018**
*Politecnico di Torino, Torino, Italy*

## PUBLICATIONS

**Journal Papers**

1. **M. Alutto**, S. Bellotti, F. Dabbene and C. Ravazzi, "Modeling and Control of Sustainable Transitions through Opinion-Behavior Coupling in Heterogeneous Networks," *IEEE Transactions on Computational Social Systems*, 2026, doi: 10.1109/TCSS.2026.3715278.
2. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, "Network Behavioral-Feedback SIR Epidemic Model," *Automatica*, vol. 193, Art. 113220, 2026, doi: 10.1016/j.automatica.2026.113220.
3. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, "On the Dynamic Behavior of the Network SIR Epidemic Model," *IEEE Transactions on Control of Network Systems*, vol. 12, no. 1, pp. 177–189, 2025, doi:

10.1109/TCNS.2024.3448136.

### Preprints

1. **M. Alutto**, L. Cianfanelli, G. Como, F. Fagnani and F. Parise, “Optimal Control for Behavioral-Feedback SIR Epidemic Model,” preprint arXiv:2512.08544, 2026.
2. **M. Alutto**, Q. Xu, F. Dabbene, H. Ishii and C. Ravazzi, “Predictive Control Strategies for Sustaining Innovation Adoption on Multilayer Social Networks,” preprint arXiv:2509.01457, 2025.

### Conference Papers

1. **M. Alutto**, L. Zino, K. H. Johansson and A. Fontan, “On Leadership Emergence in Opinion Dynamics on Social Networks,” accepted to the 65th IEEE Conference on Decision and Control (CDC 2026), preprint arXiv:2606.05825.
2. **M. Alutto**, L. Zino, K. H. Johansson and A. Fontan, “On a Co-evolving Opinion-Leadership Model in Social Networks,” Proc. of the 23rd IFAC World Congress, Busan, Republic of Korea, 2026.
3. **M. Alutto**, C. Ravazzi, F. Dabbene, K. H. Johansson and A. Fontan, “On a Coupled Adoption-Opinion Framework for Competing Innovations,” Proc. of the 23rd IFAC World Congress, Busan, Republic of Korea, 2026.
4. **M. Alutto**, Q. Xu, F. Dabbene, H. Ishii and C. Ravazzi, “Model Predictive Control for Coupled Adoption-Opinion Dynamics,” Proc. of the 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, 2025.
5. **M. Alutto**, L. Cianfanelli, G. Como, F. Fagnani and F. Parise, “Behavioral-Feedback SIR Epidemic Model: Analysis and Control,” Proc. of the 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, 2025.
6. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, “Multiple Peaks in Network SIR Models,” Proc. of the 61st IEEE Conference on Decision and Control (CDC 2022), Cancún, Mexico, 2022.
7. **M. Alutto**, G. Como and F. Fagnani, “On SIR Epidemic Models with Feedback-Controlled Interactions and Network Effects,” Proc. of the 60th IEEE Conference on Decision and Control (CDC 2021), online, 2021.

### PhD Thesis

1. **M. Alutto**, “Network Effects and Behavioral Feedback in Epidemic Models,” PhD thesis, Politecnico di Torino – Università di Torino, 2025.

## CONFERENCE PRESENTATIONS & SEMINARS

### Conference presentations

- *On a Co-Evolving Opinion-Leadership Model in Social Networks* and *On a Coupled Adoption-Opinion Framework for Competing Innovations*, 23rd IFAC World Congress, Busan, Republic of Korea, August 2026.
- *Network Effects and Behavioral Feedback in Epidemic Models*, 14th European Conference on Mathematical & Theoretical Biology (ECMTB 2026), University of Graz, Austria, July 2026 (invited minisymposium talk).
- *Behavioral-Feedback SIR Epidemic Model: Analysis and Control* and *Model Predictive Control for Coupled Adoption-Opinion Dynamics*, 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, December 2025.
- *Network Effects and Behavioral Feedback in Epidemics*, SIAM Conference on Applications of Dynamical Systems (DS25), Denver, USA, May 2025.
- *Multiple Peaks in Network SIR Epidemic Models*, 61st IEEE Conference on Decision and Control (CDC 2022), Cancún, Mexico, December 2022.
- *Multi-modal Behaviours in Network SIR Model*, 25th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2022), Bayreuth, Germany, September 2022.
- *On SIR Epidemic Models with Feedback-Controlled Interactions and Network Effects*, 60th IEEE Conference on Decision and Control (CDC 2021), online, December 2021.

### Posters

- *On Leadership Emergence in Opinion Dynamics on Social Networks*, 3rd Digital Futures Young Scientist Conference, Stockholm, Sweden, March 2026.

### Other invited talks and seminars

- Invited seminars during the ASPIRE research visit at the University of Tokyo, Kyushu University, and the RACOT Seminar, Japan, July 2025.
- Seminar hosted by prof. Karl Henrik Johansson’s group, Division of Decision and Control Systems, KTH, October 2024.

- Invited lecture in “Topics in Dynamical Systems” (MATH 7280), prof. Steven Strogatz, Cornell University, November 2023.
- FIND Graduate Day, School of Electrical and Computer Engineering, Cornell University, September 2023.

## AWARDS & MEMBERSHIPS

- Young Talent Project Scholarship, Politecnico di Torino and Fondazione CRT, 2014.
- IFAC – Affiliate since 2026.
- IEEE Technical Committee on Network Systems – Member since 2025.
- GNAMPA (Gruppo Nazionale per l’Analisi Matematica, la Probabilità e le loro Applicazioni), Istituto Nazionale di Alta Matematica – Member since 2024.
- IEEE Control Systems Society (CSS) – Member since 2023.
- IEEE – Member since 2023.
- IEEE Young Professionals – Member since 2023.

## OTHER ACTIVITIES

- Supervision of master’s theses and research internships:
  - David Puustinen (KTH, 2025–present), *“Analyzing Opinion Spreading in Dynamic Signed Social Networks Using Real-World Data,”* co-supervised with Dr. Pelin Şekercioğlu.
  - Jakob Götberg (KTH, 2026), *“Competing Technology Adoption in Networked Systems using Real-World Data,”* co-supervised with Assistant Prof. Angela Fontan.
  - Sofia Bellotti (Politecnico di Torino, 2025), *“Estimating Innovation Adoption Parameters in a Coupled Opinion–Adoption Model,”* co-supervised with prof. Chiara Ravazzi and prof. Fabrizio Dabbene.
  - Davide Sipione (Politecnico di Torino, 2024), *“On Convexity and Stability of Multicommodity Dynamical Flow Networks,”* co-supervised with prof. Giacomo Como.
- Reviewer for international journals and conference proceedings:
  - Journals (27): Automatica, IEEE Transactions on Automatic Control, IEEE Transactions on Control of Network Systems, Nonlinear Analysis: Hybrid Systems, European Journal of Control, International Journal of Control, Applied Probability Journals, Review Commons, IEEE Open Journal of Control Systems, IEEE Control Systems Letters, PLOS Computational Biology.
  - Conference proceedings (13): IEEE Conference on Decision and Control, IEEE American Control Conference, IEEE European Control Conference, IFAC World Congress.
- Member of the organizing team of the workshop “Algorithmic Game Theory, Mechanism Design, and Learning,” Politecnico di Torino, November 2022.

## SKILLS

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| *Languages* | **Italian** (native); **English** (fluent – IELTS Certification 6.5) |
| *Programming* | Python, LaTeX, C, C++, R, Matlab |